

$V \supset T(V) \supset T^2(V) \supset \dots$ (terminates in $T^\infty V = 0$ if T is nilpotent) and prove ① $\Rightarrow$ ② for nilpotent T

$$= \begin{bmatrix} \lambda_1 & * \\ 0 & Q_B^{-1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q \end{bmatrix} = \begin{bmatrix} \lambda_1 & *Q \\ 0 & Q_B^{-1} Q \end{bmatrix} !$$

Agenda from yesterday | T operator on V of dim n

① T nilpotent and diagonal $\Rightarrow T=0$

② T nilpotent $\Leftrightarrow T$ has a triangular matrix with diag all 0 $\Leftrightarrow P_T(x) = x^n \Leftrightarrow T^n = 0$

③ $P_T(x)$ factors fully $\Leftrightarrow T$ has a Δ matrix with diag entries roots of $P_T(x)$

④ Fitting decomposition of V relative to T
 $V = \ker T^\infty \oplus \text{Im } T^\infty$

$\infty = \text{sufficiently large } N$
 $(N = \dim V \text{ works!})$

Proofs ③ $\Leftarrow$ If T has matrix $\begin{bmatrix} a_{11} & * \\ 0 & a_{nn} \end{bmatrix}$ wrt some basis then $P_T(x) = \prod_{i=1}^n (x - a_{ii})$

$\Rightarrow$: Suppose $P_T(x) = (x - \lambda_1) \dots (x - \lambda_n)$

How to choose a basis to get a Δ matrix?

Let v_1 be a λ_1 -evector.

$A = \begin{bmatrix} \lambda_1 & * \\ 0 & B \end{bmatrix}$ for any completion of basis (pick one)
 matrix of T looks like this
 We want a matrix P s.t. $P^{-1}AP$ is upper Δ

$$p_T(A) = (x - \lambda_1) \cdots (x - \lambda_n) = (x - \lambda_1) p_T(B) \Rightarrow p_T(B) = (x - \lambda_2) \cdots (x - \lambda_n)$$

$$\text{b/c } xI - A = \left[\begin{array}{c|c} x - \lambda_1 & - * \\ \hline 0 & xI_{n-1} - B \end{array} \right]$$

B is the matrix of an operator on $\text{span}(v_2, \dots, v_n)$ on n

By induction, we have an $(n-1) \times (n-1)$ invertible matrix Q s.t. $Q^{-1} B Q$ is upper Δ

$$\text{Define } P = \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array} \right] \text{ Then } P^{-1} = \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array} \right]$$

$$P^{-1} A P = \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array} \right] \left[\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & B \end{array} \right] \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array} \right]$$

Rmk Can also analyze decreasing chain

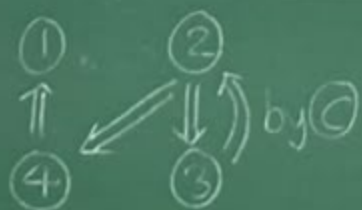
$V \supset T(V) \supset T^2(V) \supset \dots$ (terminates in $T^\infty V = 0$ if T is nilpotent)
and prove ① $\Rightarrow$ ② for nilpotent T

$$= \left[\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1} B \end{array} \right] \left[\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q \end{array} \right] = \left[\begin{array}{c|c} \lambda_1 & * Q \\ \hline 0 & Q^{-1} B Q \end{array} \right] !$$

- (i) "not enough evalules" ie $p_T(x)$ does not factor fully (defect of F)
 (ii) "not enough evectors for a given evalue" $\dim V > \sum \dim \text{of spaces}$

$\dim V = \sum_{\lambda \in F} \dim \ker (T - \lambda I)$ spaces
 For now we will see how to tackle situation (ii), assuming $p_T(x)$ factors fully.

Pf of B



$$v \in \ker(T^k) \Leftrightarrow T^k v = 0$$

$$v \in \ker(T^{k+1}) \Leftrightarrow T^{k+1} v = 0$$

Take basis of $\ker T$

Add to get basis of $\ker(T^2)$

Continue ...

Matrix of $\ker(T^\infty)$ w/ this basis
 is upper Δ

Left to do: $(1) \Rightarrow (2)$ Pf For any operator T on V
 observe: Increasing chain of subspaces

$$0 \subset \ker T \subset \ker(T^2) \subset \dots \quad (\text{If } T \text{ nilpotent eventually we get } \ker(T^N) = V)$$

If $\ker(T^k) = \ker(T^{k+1})$ then $\ker(T^k) = \ker(T^{k+j}) \quad \forall j \geq 0$.
 Write a precise pf by following your nose.

If T is nilpotent, $\ker(T^{\text{large}}) = V$. As $\dim \ker(T^i)$ have to grow at each $i \rightarrow i+1$ and must reach n , # steps needed to reach n is $\leq n$

$$T \text{ nilp} \Rightarrow T^{\dim V} = 0 \quad \text{So } (1) \Rightarrow (4)$$

Analysis of operator T on f.dim V -space V over field F

λ is an eval of $T \iff \ker(\lambda I - T) \neq 0 \iff \det(\lambda I - T) = 0$ i.e. λ a root of char poly

Recipe to test diagonalizability of T T diag $\iff V$ has an eigenbasis corresp to T

- ① Find char poly $p_T(x) = \det(xI - T)$. Find its roots

② For each root λ , find a basis for $\ker(\lambda I - T)$

$\iff \left\{ \begin{array}{l} \text{① } p_T(x) \text{ factors fully } \underline{\text{AND}} \\ \text{② } V = \bigoplus_{\lambda \in F} \ker(\lambda I - T) \end{array} \right.$

$\updownarrow \rightsquigarrow$ Uses indep of spaces
 $\dim V = \sum_{\lambda \in F} \dim \ker(\lambda I - T)$

T can fail to be diagonal for two reasons.

(i) "not enough evals" i.e. $p_T(x)$ does not factor fully (defect of F)

(ii) "not enough evectors for a given eval" $\dim V > \sum \dim \text{of spaces}$

For now we will see how to tackle situation (ii), assuming $p_T(x)$ factors fully.